Demonstration of Network Topologies and Transmission Media Using Cisco Packet Tracer

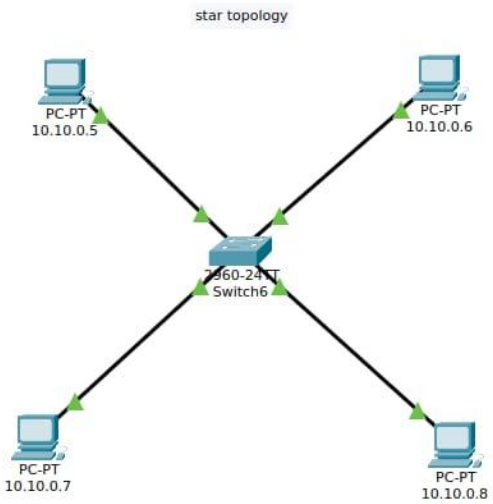
1. Introduction

Network topologies define the geometric or logical arrangement of nodes and links in a computer network. Transmission media refer to the physical or wireless channels through which data packets are transmitted between nodes. This guide provides step-by-step instructions, configurations, and expected results for demonstrating various topologies and transmission media inside **Cisco Packet Tracer**.

2. Common Network Topologies in Packet Tracer

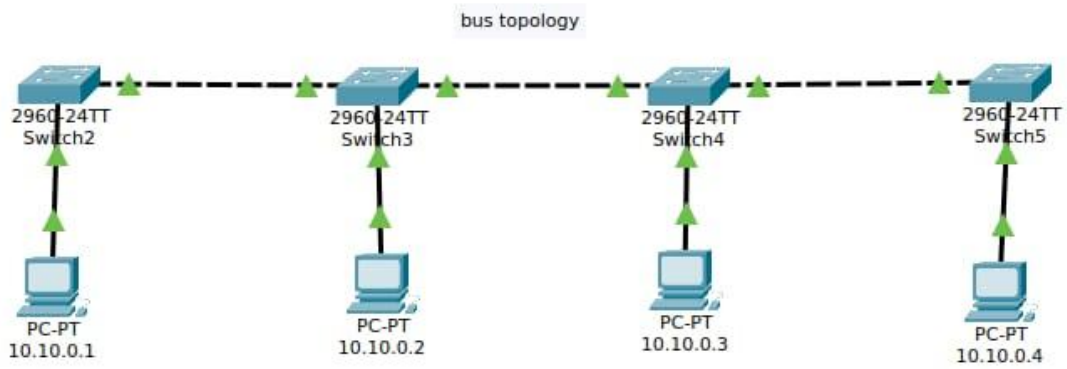
A. Star Topology

- **Description:** All end devices (PCs, laptops) are connected to a central hub or switch. If one cable fails, only that device is disconnected from the network.
- **Packet Tracer Implementation Steps:**
    1. Place a central **Switch** (e.g., 2960-24TT).
    2. Place 4 **End Devices** (End Devices > PC-PT).
    3. Connect each PC to the switch using **Copper Straight-Through cables** (Connections > Copper Straight-Through).
    4. Assign static IP addresses to each PC (e.g., 192.168.1.10 through 192.168.1.13 with subnet mask 255.255.255.0).



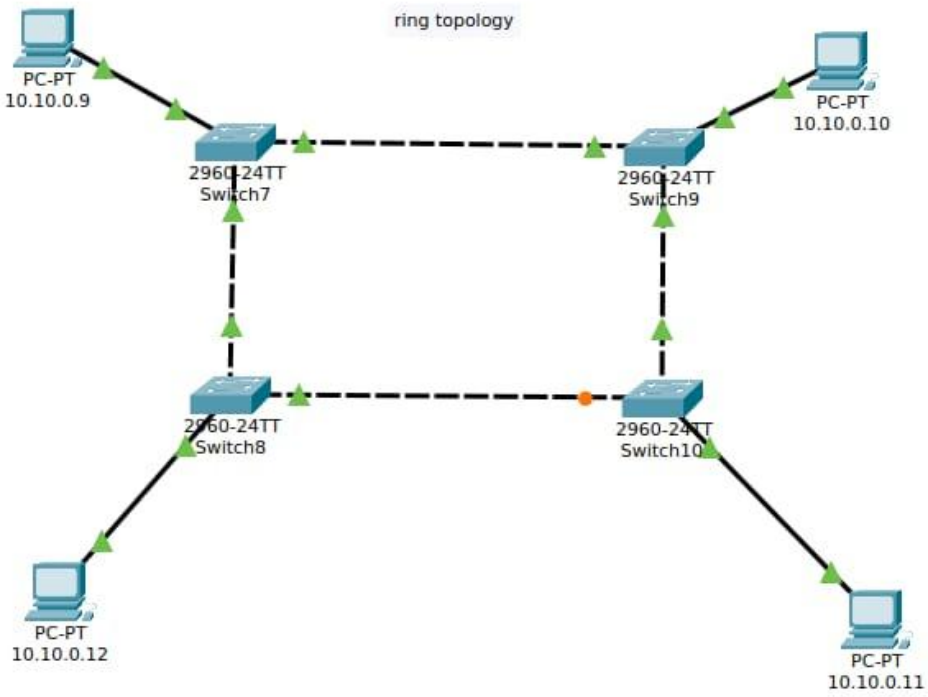
B. Bus Topology

- **Description:** Devices are connected to a single central backbone cable (coaxial cable structure).
- **Packet Tracer Implementation Steps:**
    1. In modern Packet Tracer, traditional BNC coaxial bus networks are simulated by linking multiple switches or using hubs in a linear daisy-chain.
    2. Place 3 **Hubs** (Hub-PT) and connect them in a linear sequence using **Copper Crossover** or **Straight-Through** cables.
    3. Connect PCs to each hub.



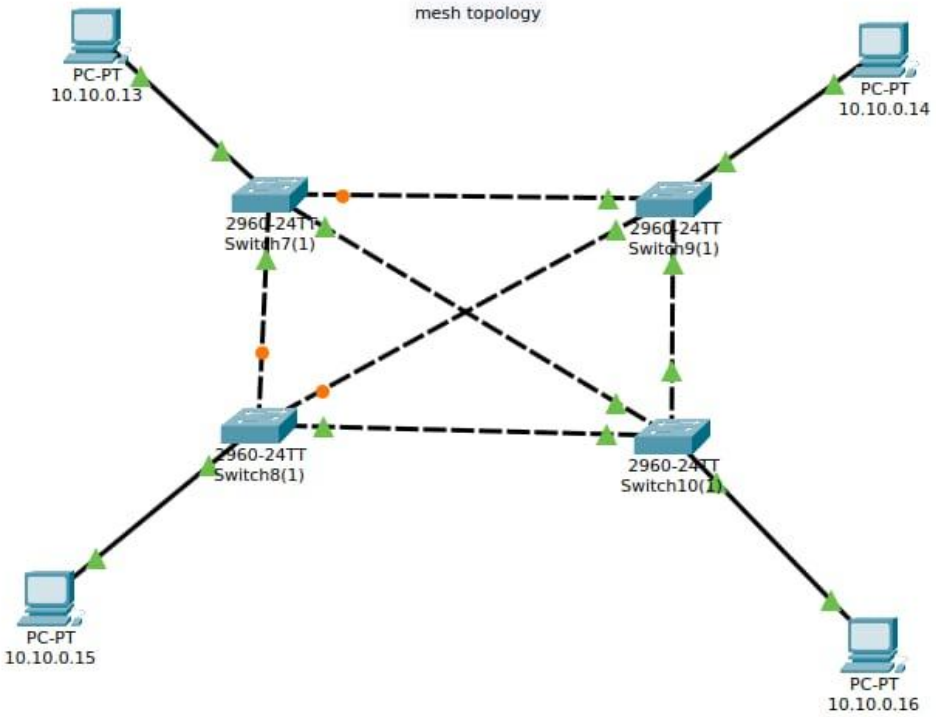
C. Ring Topology

- **Description:** Devices are connected in a circular loop where data travels in one direction (or dual directions) from node to node.
- **Packet Tracer Implementation Steps:**
    1. Place 4 **Switches** (2960).
    2. Connect Switch 0 to Switch 1, Switch 1 to Switch 2, Switch 2 to Switch 3, and Switch 3 back to Switch 0 using **Copper Crossover cables**.
    3. Attach a PC to each switch.



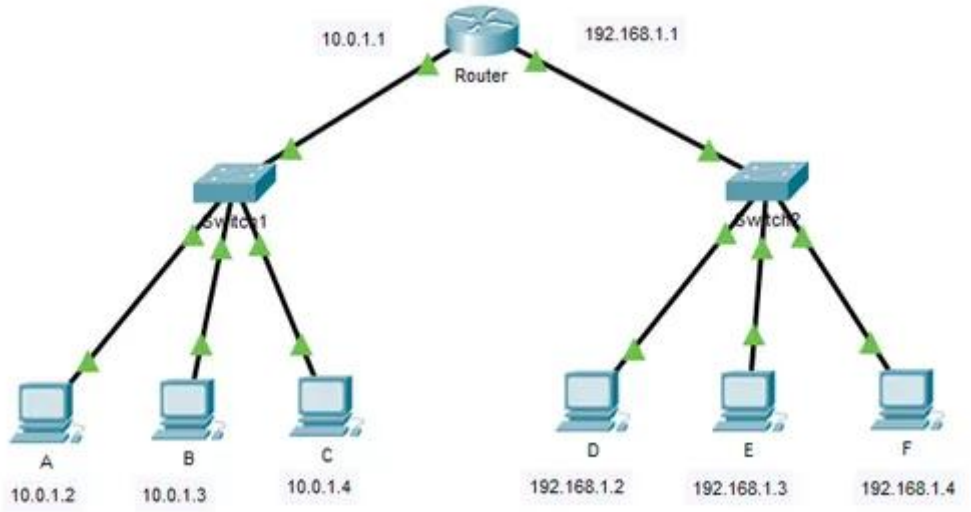
D. Mesh Topology

- **Description:** Every node is connected to every other node, providing maximum redundancy and fault tolerance.



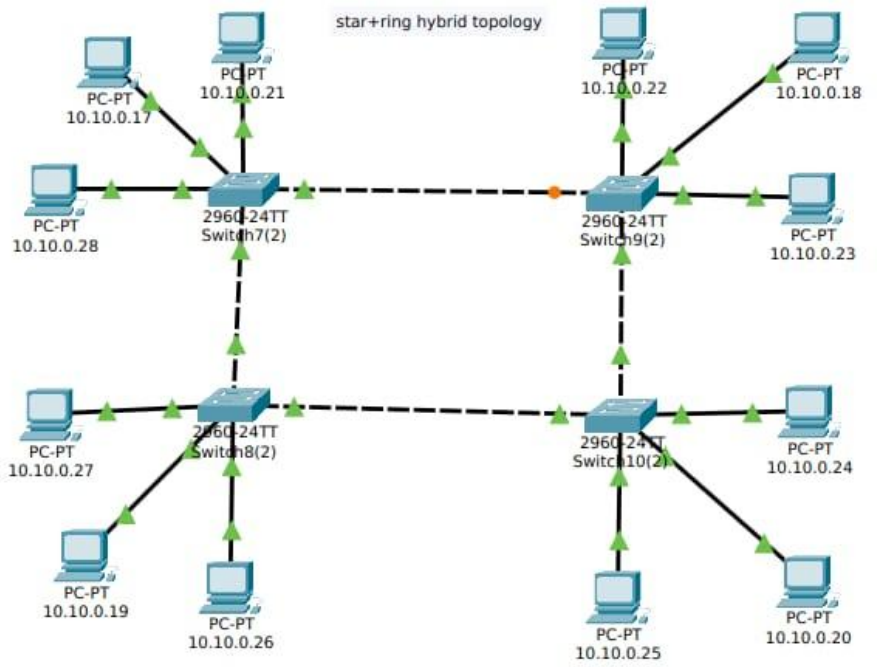
E. Tree / Hierarchical Topology

- **Description:** A combination of star and bus topologies arranged in a hierarchy (levels/branches), typical of enterprise campus networks.
- **Packet Tracer Implementation Steps:**
    1. Place a Core **Router** at the top.
    2. Connect Distribution **Switches** to the router.
    3. Connect Access **Switches** and end devices underneath the distribution switches.



F. Hybrid (Star-Ring) Topology

- **Description:** A hybrid topology combines two or more different standard topologies. In a Star-Ring hybrid, multiple **Star** networks (where end devices connect to a central switch) are linked together in a **Ring** backbone configuration via inter-switch connections.
- **Packet Tracer Implementation Steps:**
    1. Place 3 **Switches** (e.g., 2960-24TT) on the workspace to form the ring backbone.
    2. Connect the switches in a circular loop using **Copper Crossover cables** (Switch0 to Switch1, Switch1 to Switch2, and Switch2 to Switch0).
    3. Attach multiple **PCs** to each individual switch using **Copper Straight-Through cables**, creating a local Star topology around every switch.
    4. Assign static IP addresses within the same subnet (e.g., 192.168.1.x with subnet mask 255.255.255.0).



3. Transmission Media Demonstration

Packet Tracer categorizes transmission media under the **Connections** menu (represented by the lightning bolt icon):

Media Type	Packet Tracer Icon / Cable Name	Typical Use Case
Copper Straight-Through	Solid black line	Connecting dissimilar devices (e.g., PC to Switch, Router to Switch)
Copper Crossover	Dashed red line	Connecting similar devices directly (e.g., PC to PC, Switch to Switch, Router to Router)
Fiber Optic	Orange solid line	Long-distance, high-speed connections between switches or routers
Coaxial	Black coaxial cable	Connecting legacy cable modems, TVs, or cloud modules
Wireless (Radio	Wavy red line /	Connecting laptops/smartphones wirelessly via Wi-

Media Type	Packet Tracer Icon / Cable Name	Typical Use Case
Frequency)	Access Point	Fi (PT-HOST or Wireless-N Router)